

System Concept — PD-RPM Remote Care System

The concept. PD-RPM is a monitoring and decision-support layer wrapped around therapy the patient already performs at home. It does not dialyze anyone and it does not control the cycler. It observes each completed treatment, adds the two things the cycler cannot see — the patient's vitals and the patient's symptoms — and delivers a reviewed signal to the person able to act on it. The diagram below shows the four functional blocks and the boundary. Home acquisition (A1–A4) captures data; transport (T1) moves it; the monitoring center (M1–M5) turns readings into a ranked queue and, when warranted, a page; the care team (U1–U3) decides.

The commitment the diagram makes. No automated element crosses into clinical decision-making. M3 may flag and M5 may escalate, but every path terminates at a human, and the only arrow returning into the system carries a decision a clinician already made. Gray dashed elements lie outside the boundary: PD-RPM defines an interface to the cycler and the EHR but builds neither.

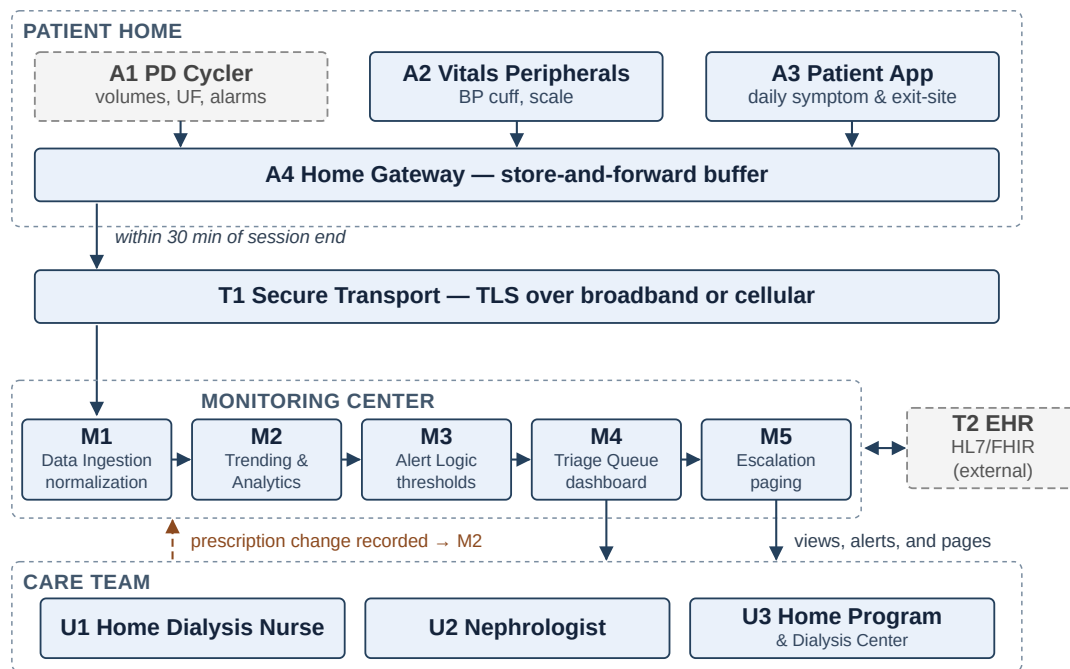


Figure 1. PD-RPM system concept. Solid arrows carry data; the brown dashed arrow is the clinician feedback path; gray dashed boxes lie outside the system boundary.

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| ① Declining ultrafiltration | A1 → A4 → T1 → M1 → M2 → M3 → M4 → U1 → U2 → M2 |
| ② Early peritonitis | A3 → A4 → T1 → M1 → M2 → M3 → M4 → M5 → U1 → U3 |
| ③ Supply shortfall | A1 → A4 → T1 → M1 → M2 → M3 → M4 → M5 → U3 |

① **A declining ultrafiltration trend.** A 58-year-old man, fourteen months on automated PD, works full time, comfortable with a smartphone, living alone in a suburb with reliable broadband and no care partner. Over three weeks his net ultrafiltration drifts downward without any single night looking abnormal. A1 reports fill and drain volumes nightly, A4 forwards them through T1, M1 normalizes them, and M2 finds the seven-day mean twenty-four percent below the thirty-day mean. M3 raises a non-urgent flag, M4 places it in the nurse's queue, and the nephrologist shortens the dwell. That change is recorded back into M2 so the following fourteen days can be judged against it rather than against memory.

② **Early peritonitis.** A 71-year-old woman, four months on continuous ambulatory PD, mild cognitive impairment, her daughter as care partner, a rural home on cellular service only. She reports cloudy effluent and abdominal discomfort on the daily sixty-second check in A3. Coverage is out that evening, so A4 buffers and forwards when the link returns. M1 ingests the entry, M3 recognizes a positive peritonitis screen and marks it urgent, which promotes it to the head of the M4 queue and triggers M5 to page the on-call nurse within fifteen minutes. U3 brings her in for an effluent cell count the same day. The value delivered here is lead time, not data.

③ **A supply shortfall.** A 44-year-old man, three years on automated PD, spouse as care partner, a fourth-floor urban apartment with storage for roughly two weeks of solution. He does nothing at all in this scenario, which is the point. A1 reports cassette and bag consumption with each session, M1 normalizes it, and M2 projects days of supply remaining from observed consumption rather than from order history. When the projection falls below fourteen days the supply rule in M3 fires and M5 notifies the home program, which ships early. Because the projection follows what was used rather than what was ordered, it does not inherit the ordering distortions examined in Module 2.

Coverage. Together the three scenarios exercise every operational need from M3A1: ON-1 and ON-4 in the first, ON-3 in the second, ON-5 in the third, and ON-2 throughout, since no path asks the patient for more than a sixty-second check.